DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT



Calculus and Optimization

Limit, Continuity and differentiablity

Lecture No. 02



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

BASICS of LIMITS & CONTINUITY

(PART-1)

Topics to be Covered



Topic

BASICS of LIMITS & CONTINUITY

(Part-2)

BASICS of LIMITS

Infinity → Infinity is Not a very large Number. It is the presentation of that Concept which is beyond Imagination.

→ & Infinity is Not unique, it depends upon the Imagination capacity of an Individual.

→ $\frac{\text{Something}}{\infty} \cong 0$ Assumption

$\frac{\text{Something}}{0} = \text{N.D.}$ (FACT)

GOD GOD GOD

$\infty + \infty = \infty$ ≈ N.D

I don't know — you don't know — we don't know

$\infty \times \infty = \infty$ ≈ N.D

$\infty^{\infty} = \infty$ ≈ N.D

$\infty - \infty = ?$ = IND form.

$\frac{\infty}{\infty} = ?$ = IND form.

$\infty^{0} = ?$ = IND form.

$1^{\infty} = ?$ = IND form.

$0 \times \infty = ?$ = IND form.

$0^{0} = ?$ = IND form.

$\frac{0}{0} = ?$ = IND form.

Note: if final ans is ∞ or $-\infty$ then we say that it is **Not Defined**

But if final ans is $\frac{0}{0}$ then we say that it is **IND form**

PROPERTIES of LOG → (let $a>1$)

① $\log_a x = b \Rightarrow x = a^b$

eg $\log_4 64 = ? = 3 \Rightarrow 64 = 4^3$

eg $\log_5 625 = ? = 4 \Rightarrow 625 = 5^4$

eg $\log_a 1 = 0$, eg $\log_a 0 = -\infty$ where $a>1$

$\left(\because a^{-\infty} = \frac{1}{a^{\infty}} = \frac{1}{\infty} = 0\right)$

eg $\log_a \infty = \infty$ eg $\log_a a = 1$

② $\log_a x > \log_a y \Rightarrow x > y$

Provided $a>1$ i.e when Base is greater than 1, $\log_a x$ is an Increasing func^n.

Domain $= (0, \infty)$

Range $= (-\infty, \infty)$



③ Product formula :

$$\log_a(x \cdot y) = \log_a x + \log_a y$$

④ Quotient formula –

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

⑤ Power formula –

$$\log_a(x^y) = y \log_a x$$

⑥ Base change formula –

$$\log_y x = \frac{\log_a x}{\log_a y},$$

$$\Rightarrow \log_\infty x = \frac{\log e^x}{\log e^\infty} = \frac{\log e^x}{\infty} = 0$$

$$\Rightarrow \log_1 1 = \frac{\log e^1}{\log e^1} = \frac{0}{0} = ? = \text{N.D}$$

Note – $\log_1 x = \frac{\log e^x}{\log e^1} = \frac{\log e^x}{0} = \text{N.D.}$ (i.e. $\log_1 x$ is not defined)

(7) Reciprocal formula :-

$$\log_y x = \frac{1}{\log_x y}$$

(8) $\log_{a^b}(x) = \frac{1}{b}\log_a x$

(9) $a = e^{\log_e a}$

(10) $a^n = e^{\log_e (a^n)} = e^{n(\log_e a)}$

(11) $a^{\log_e b} = b^{\log_e a}$

INDETERMINATE form :- There are exactly 7 Indeterminate forms for eg $\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty, 0^0, \infty^0, 1^\infty$

" If any Mathematical Expression has Multiple Answers then that expression is said to be in INDETERMINATE form "

L-Hospital Rule :- this Rule provides unique answer out of ∞ answers & this Rule is applicable only for $\frac{0}{0}$ and $\frac{\infty}{\infty}$ form.

Rule: Separately Differentiate N^r & D^r, equal number of times, until we are free from $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

i.e $\lim_{x\to a}\left\{\frac{f(x)}{g(x)}\right\} \simeq \frac{0}{0} = \lim_{x\to a}\frac{f'(x)}{g'(x)} \simeq \frac{0}{0} = \lim_{x\to a}\frac{f''(x)}{g''(x)} = \cdots\cdots = \text{unique ans}$

eg $\lim_{x\to\infty}\left(\frac{x^3}{e^x}\right) = \frac{\infty}{\infty}\text{ form} = \lim_{x\to\infty}\left(\frac{3x^2}{e^x}\right) = \frac{\infty}{\infty} = \lim_{x\to\infty}\left(\frac{6x}{e^x}\right) = \frac{\infty}{\infty} = \lim_{x\to\infty}\left(\frac{6}{e^x}\right) = \frac{6}{e^\infty}$

$= \frac{6}{\infty} = 0$ i.e unique ans exist.

Note:- limit exist means "There exist unique, constant value that must be free from x"

eg $\frac{0}{0} = 1, 2, 3, \cdots, -1, -2, -3, \cdots, \frac{1}{2}, \frac{3}{4}, \sqrt{5}, \frac{-1}{7}$ ---------

Multiple Answers exist.

eg $\frac{\infty}{\infty} = ? = \frac{1/0}{1/0} = \frac{0}{0}$ = Multiple Ans exist.

eg $0 \times \infty = ? = 0 \times \frac{1}{0} = \frac{0}{0}$ = Multiple Ans exist.

eg $\infty - \infty = ?$ = Have patience, I will discuss later

eg $0^0 = ? = k$ (let)

i.e. $k = 0^0$

$\log_e k = \log(0^0)$

$= 0 \times \log(0)$

$= 0 \times (-\infty)$

$= 0 \times \left(\frac{-1}{0}\right)$

$\log_e k = \frac{0}{0}$

$k = e^{\frac{0}{0}}$ = Multiple Ans exist.

eg $\infty^0 = ? = k$ (let)

i.e. $k = \infty^0$

$\log_e k = \log(\infty^0)$

$= 0 \times \log \infty$

$= 0 \times (\infty)$

$= 0 \times \left(\frac{1}{0}\right)$

$\log_e k = \frac{0}{0}$

i.e. $k = e^{\frac{0}{0}}$ = Again Multiple answer exist.

Eg $1^{\infty} = ? = K$ (let)

is $K = 1^{\infty}$

$\log_e K = \log(1^{\infty})$

$= \infty \times \log(1)$

$= \infty \times (0)$

$= \frac{1}{0} \times 0$

$\log_e K = \frac{0}{0}$

$K = e^{\frac{0}{0}}$ = Multiple Ans exist.

Note: $1^{20} = 1$, $1^{50} = 1$, $1^{200} = 1$, But $1^{\infty} \neq 1$

Doubt: Check the Nature of

① 1^{20}, ② 0^{∞} ③ ∞^{∞}

$K = 1^{20}$

$\log K = \log 1^{20}$

$= 20 \times \log 1$

$= 20 \times 0$

$\log_e K = 0$

$K = e^0 = 1$

is Determinate

Let $K = 0^{\infty}$

$\log K = \log 0^{\infty}$

$= \infty \times \log 0$

$= \infty \times (-\infty)$

$\log_e K = -\infty$

$K = e^{-\infty} = 0$

is Determinate

$K = \infty^{\infty}$

$\log K = \log \infty^{\infty}$

$= \infty \times \log \infty$

$= \infty \times \infty$

$\log_e K = \infty$

$K = e^{\infty} = \infty$

is N.D.

Neighbourhood of Real Number 'a' ⟶ means we are considering an open interval of length $2h$ with centered at Point 'a'

ie Nbd of $a = (a-h, a+h)$ where $h > 0$ & h depends on our choice. & h is very small +ve Number.

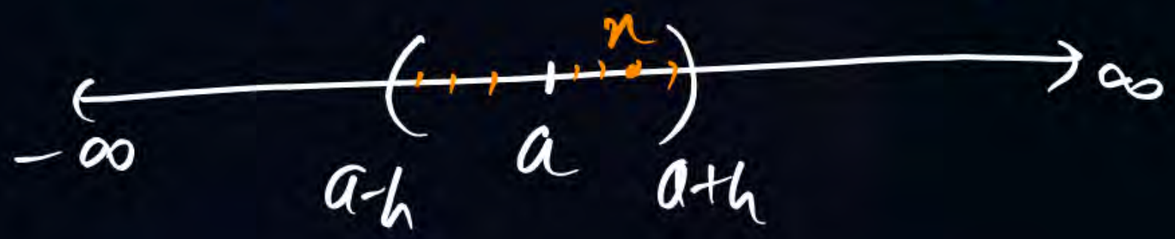
Note: if x lies in the Nbd of $a \Longrightarrow x \in (a-h, a+h)$

or

$a-h < x < a+h$

or

$x \to a$



for eg: if $x \to 3 \Longrightarrow x$ lies in the Nbd of 3 ie $x \in (2.999, 3.001)$

Concept of limit →

① $f(x) = \dfrac{x^2-9}{x-3}$

$f(x) = \dfrac{(x-3)(x+3)}{(x-3)}$

$f(3) = \dfrac{(3-3)(6)}{(3-3)} = \dfrac{0\times 6}{0} =$ DNE

i.e Exact Value at $x=3$ DNE

LHL $= 6$ i.e Approx Value of $f(x)$ in the left Nbd of 3

RHL $= 6$ i.e Approx Value of $f(x)$ in the Right Nbd of 3

$\lim\limits_{x\to 3} f(x) = 6$ (Approx Value of $f(x)$ in the Nbd of 3)

$LHL = f(2.999) = \dfrac{-0.001\times 5.999}{-0.001} = 5.999 \approx 6$

$RHL = f(3.001) = \dfrac{0.001\times 6.001}{0.001} = 6.001 \approx 6$

(2) $f(x)=\frac{|x-3|}{x-3}$

$$=\begin{cases}\frac{-(x-3)}{x-3}, & x-3<0\\ \text{DNE}, & x-3=0\\ \frac{+(x-3)}{x-3}, & x-3>0\end{cases}$$

$$=\begin{cases}-1, & x<3\\ \text{DNE}, & x=3\\ 1, & x>3\end{cases}$$

Exact Value of $f(x)$ at $x=3$ will be

$f(3)=$ DNE

Approx Value of $f(x)$ in the left Nbd of $x=3$ will be

$LHL=-1$ (exist)

Approx Value of $f(x)$ in the Right Nbd of $x=3$ will be $RHL=+1$ (exist)

$\because LHL\neq RHL \Rightarrow$ limit = DNE

i.e. App Value of $f(x)$ in the Nbd of 3 = DNE

③ $f(x)=\frac{1}{x-3}$ then $f(3)=\frac{1}{3-3}=\frac{1}{0}=$ N.D

ie exact Value at $x=3$ DNE.

$$LHL(\text{at } x=3)=\lim_{x\to 3-h} f(x)=\lim_{x\to 3-h}\left(\frac{1}{x-3}\right)=\lim_{h\to 0}\left(\frac{1}{3-h-3}\right)$$

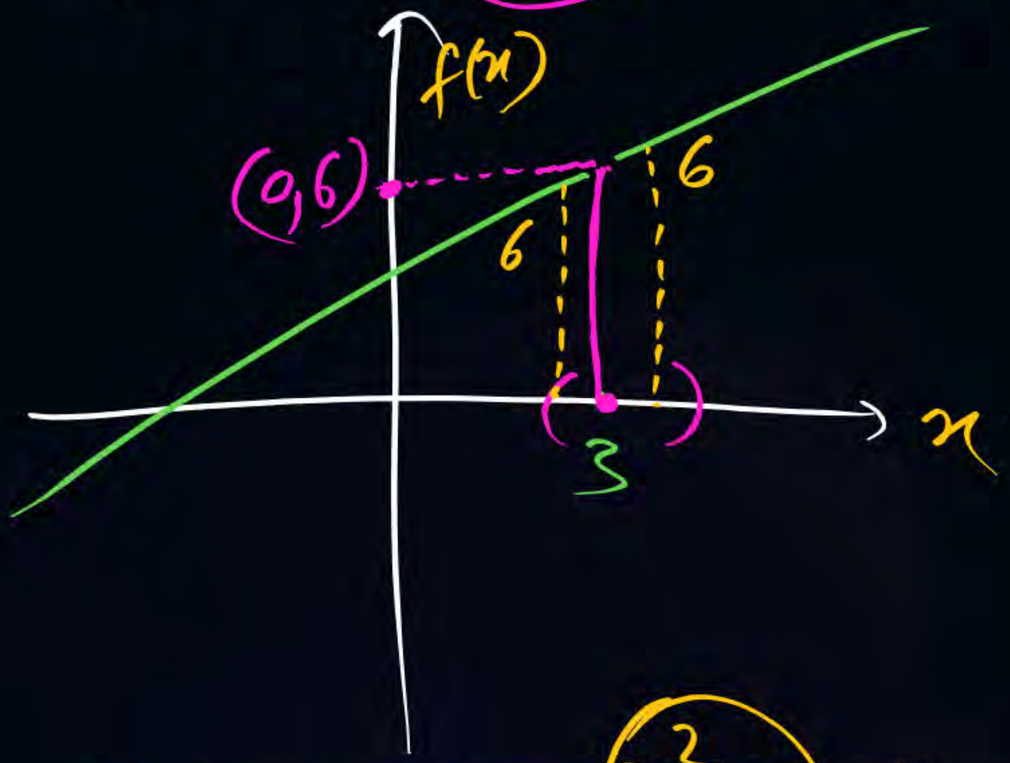
$$=\lim_{h\to 0}\left(-\frac{1}{h}\right)=-\frac{1}{0}=-\infty \text{ ie } LHL=DNE$$

$$RHL(\text{at } x=3)=\lim_{x\to 3+h} f(x)=\lim_{x\to 3+h}\left(\frac{1}{x-3}\right)=\lim_{h\to 0}\left(\frac{1}{3+h-3}\right)=\lim_{h\to 0}\left(\frac{1}{h}\right)$$

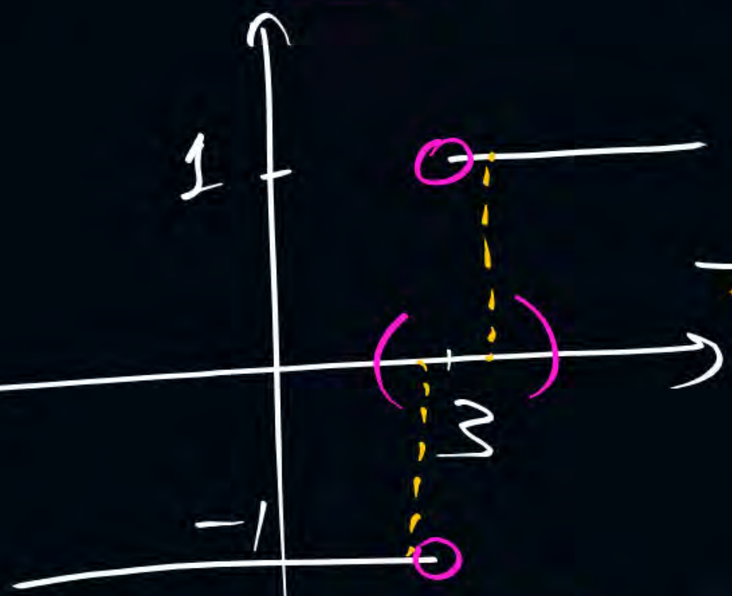
$$=\frac{1}{0}=\infty \text{ ie } RHL=DNE$$

Hence overall $\lim_{x\to 3} f(x)=DNE$

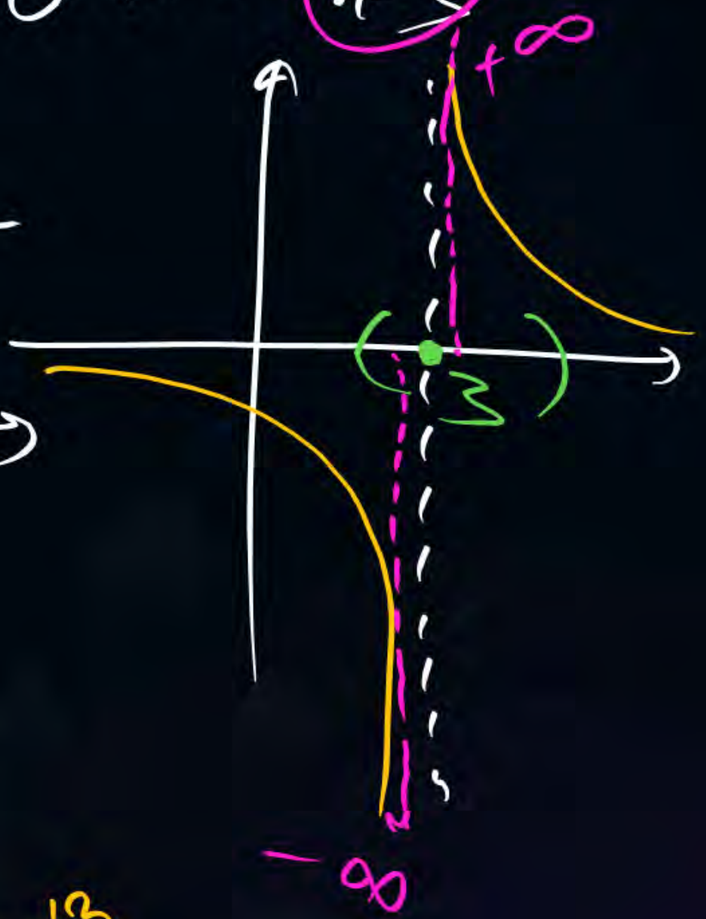
eg $f(x)=\frac{x^2-9}{x-3}$



eg $f(x)=\frac{|x-3|}{x-3}$



eg $f(x)=\frac{1}{x-3}$



Note $f(x)=\frac{x^2+9}{x-3}$ then

$f(2)=\frac{4+9}{2-3}=-13$

$\lim_{x\to 2} f(x)=\left(\frac{2^2+9}{2-3}\right)=-13$

Def^n of limit $\Rightarrow$ $\lim_{x \to a} f(x) = l \Rightarrow$ whenever x lies in the Nbd of 'a' $f(x)$ lies in the Nbd of l

eg $\lim_{x \to 0} \left(\frac{\sin x}{x}\right) = 1 \Rightarrow$ whenever x lies in the Nbd of '0' $\left(\frac{\sin x}{x}\right)$ lies in the Nbd of 1

Continuity $f(x)$ is said to be Continuous at $x = a$ if

$\lim_{x \to a} f(x) = f(a) \Rightarrow$ whenever x lies in the Nbd of a $f(x)$ lies .. .. of $f(a)$

App Value = Exact Value

i.e $(LHL = RHL) = f(a)$

Sp. Note: To check Discontinuity of $f(x)$ at $x=a$ we have 3 Methods;

(M-I) If Exact Value = DNE then $f(x)$ is Discontinuous

(M-II) if Approx Value = DNE then " " "

(M-III) if Both exist But not equal then $f(x)$ is Discont.

ie $\lim_{x \to a} f(x) \neq f(a) \Rightarrow f(x)$ is Discont at $x=a$

Q: Check which of the following fun^{n} is Cont & Discont at $x=0$

① $f(x)=\frac{\sin x}{x}$

$\because f(0)=DNE$

i.e exact value at $x=0$ DNE

$\therefore f(x)$ is Discont (by M-I)



② $f(x)=\begin{cases}\frac{\sin x}{x}, & x\neq 0\\ 1, & x=0\end{cases}$

Exact Value $=f(0)=1$ (given)

App value $=\lim_{x\to 0}\left(\frac{\sin x}{x}\right)=1$

$\because$ App Value = Exact Value $\therefore$ Cont at $x=0$.



ⓐ Both Cont

ⓑ Both Discont

ⓒ 1st Cont & 2nd Disc

ⓓ 1st Disc & 2nd Cont

Method of Solving Questions →

(M-I) By Direct Substitution (Best Method).

(M-II) By factorisation

(M-III) By Rationalisation

(M-IV) By using IND form Concept

(M-V) By using Standard Results

THANK - YOU

Tel: drpuneet Sir pw